

DATASHEET

Variable Speed Drives



Product coding : CFW500B31P0T4DB66DS
 Product reference : 14978573
 Accessory module (control) : CFW500
 :

Power supply : 380-480 V
 Input minimum-maximum voltage :
 - In :
 - Out :

Supply voltage range	380-480 V	
Overload cycle	Normal Overload (ND)	Heavy Overload (HD)
Rated current (HD)	31	31
CORRENTE_SOBRECARGA_60S_CFW		
CORRENTE_SOBRECARGA_3S_CFW		

Maximum applicable motor:

Voltage/Frequency	Power (HP/kW) [1]	
	Normal Overload (ND)	Heavy Overload (HD)

Accessory module (control) :
 Dynamic braking [2] : Standard with braking
 External electronic supply 24Vcc : Not available
 Safety Stop : Not available
 Internal RFI filter : Without filter
 :
 Link Inductor :
 Memory card : Not included in the product
 USB port : Only with plug-in
 Line frequency : 50/60Hz
 Line frequency range (minimum - maximum) : 48-62 Hz
 Phase unbalance : Less or equal to 3% of input rated line voltage
 Transient voltage and overvoltage : Category III
 Single-phase input current [3] : Not applicable
 Three-phase input current [3] : 37,8 A
 Power factor :
 Displacement factor : 0,98
 Rated efficiency :
 Maximum connections (power up cycles - on/off) per hour : 10 (1 each 6 minutes)
 DC power supply :
 Standard switching frequency :
 Selectable switching frequency : 2,5 and 15 kHz
 Real-time clock : Not available
 COPY Function : Yes, by MMF
 Dissipated power:

Mounting type	Overload	
	ND	HD
Surface		
Flange	Not applicable	Not applicable

Source available to the user

Output voltage :
 Maximum capacity :
 Power supply : Switched-mode power supply
 Control method : V/f, VVW, Sensorless and Encoder
 Encoder interface :
 Control output frequency : 0-500 Hz
 Frequency resolution : 0,015 Hz
 - Speed resolution : 1% of rated speed
 - Speed range : 1:20
 - Speed resolution : 1% of rated speed
 - Speed range : 1:30
 - Speed resolution : 0,5% of rated speed
 - Speed range : 1:100
 - Speed resolution : 0,1% of nominal speed
 - Speed range : Up to 0 rpm
 Quantity (standard) :

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Levels	:
Impedance for voltage input	:
Impedance for current input	:
Function	:
Maximum allowed voltage	:
Quantity (standard)	:
Activation	:
Maximum low level	:
Minimum high level	:
Input current	:
Maximum input current	:
Function	:
Maximum allowed voltage	:
Analog outputs	
Analogic outputs - Quantity (standard)	:
Levels	:
RL for voltage output	:
RL for current output	:
Function	:
Digital outputs	
Digital outputs - Quantity (standard)	:
Maximum voltage	:
Maximum current	:
Function	:
-	:
-	:
-	:
-	:
-	:
-	:
-	:
-	:
-	:
Available protection	
- Output phase-phase overcurrente/Short	
- Overcurrent/Short circuit phase-ground	
- Under/Overtoltage in power	
- Heat sink overtemperature	
- Motor overload	
- IGBT's modules overload	
- Fault/External alarm	
- Programming error	
Operation interface (HMI)	
Avaliability	:
Installation	:
Number of HMI buttons	:
Display	:
Indication accuracy	:
Speed resolution	:
Standard HMI degree of protection	:
HMI battery type	: Not applicable
HMI battery life expectancy	: Not applicable
Remote HMI type	: Accessory
Remote HMI frame	: Not applicable
Remote HMI degree of protection	: IP54
Enclosure	: IP66
Degree of pollution	: 2
RoHS	: Yes
Conformal Coating	:
- Size	: 00009
- Height	:
- Width	:
- Depth	:
- Weight	:
Mechanical Installation	
Mounting position	:

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Mechanical Installation

Fixing screw :
 Tightening torque :
 Allows side-by-side assembly :
 Minimum spacing around the inverter:
 - Top :
 - Bottom :
 - Front :
 - Side :

Cable gauges and tightening torques:

	Recommended cable gauge	Recommended tightening torque
Power		
Braking		
Grounding		
Control		

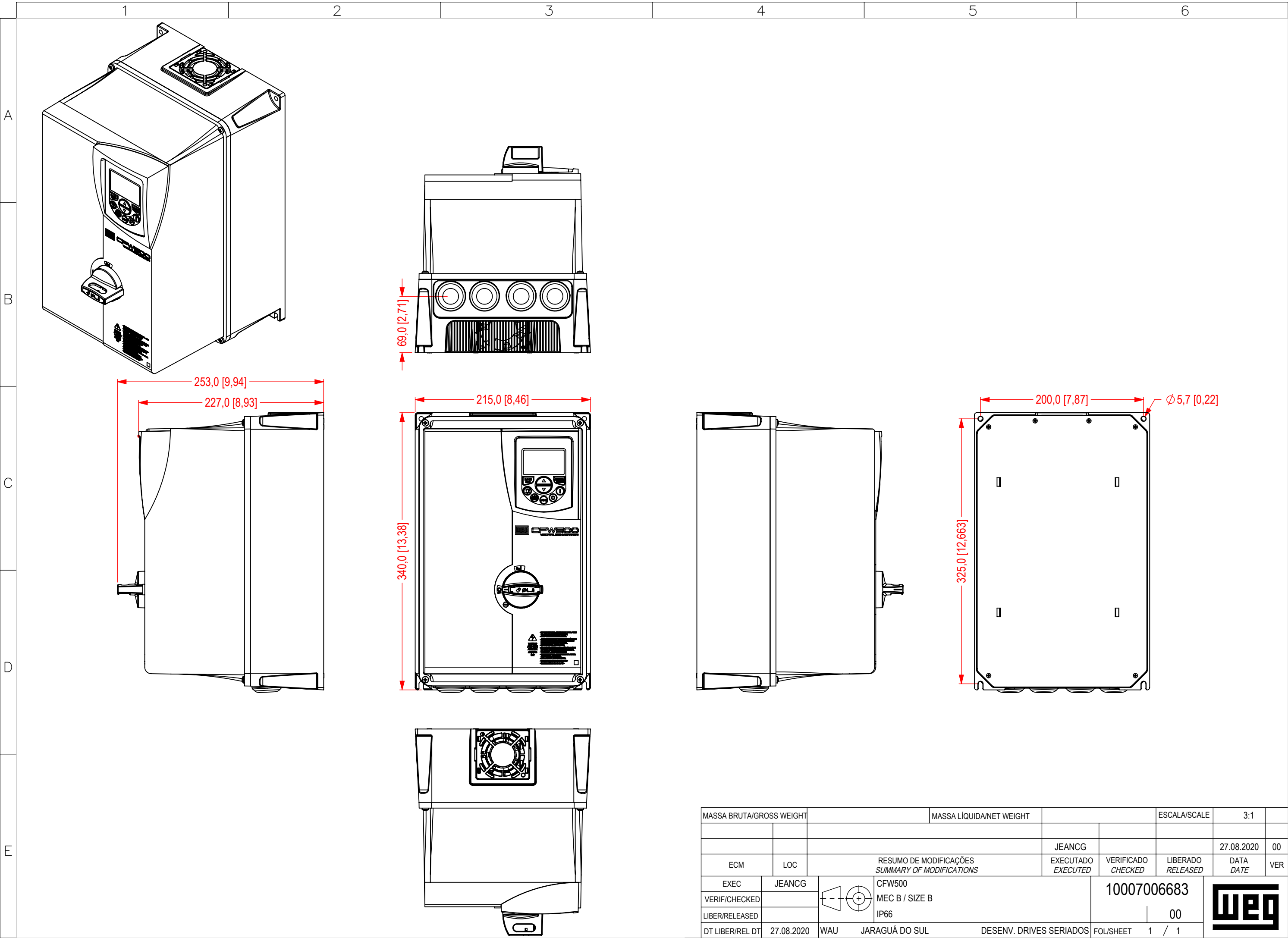
SoftPLC : Yes, incorporated
 Maximum breaking current :
 Minimum resistance for the brake resistor :
 Recommended aR fuse :
 Recommended circuit breaker :
 Disconnect switch : With disconnect switch
 Motor coupling box : Not applicable

Standards

Safety	- UL 508C - Power conversion equipment. - UL 840 - Insulation coordination including clearances and creepage distances for electrical equipment. - EN 61800-5-1 - Safety requirements electrical, thermal and energy. - EN 50178 - Electronic equipment for use in power installations. - EN 60204-1-Safety of machinery. Electrical equipment of machines. Part 1: General requirements. Note: To have a machine in accordance with that standard, the manufacturer of the machine is responsible for the installation of an emergency-stop device and a network switching equipment. - EN 60146 (IEC 146) - Semiconductor converters. - EN 61800-2 - Adjustable speed electrical power drive systems - Part 2: General requirements - Rating specifications for low voltage adjustable frequency AC power drive systems.
Electromagnetic Compatibility	- EN 61800-3 - Adjustable speed electrical power drive systems - Part 3: EMC product standard including specific test methods. - EN 55011 - Limits and methods of measurement of radio disturbance characteristics of industrial, scientific and medical (ISM) radio-frequency equipment. - CISPR 11 - Industrial, scientific and medical (ISM) radio-frequency equipment - Electromagnetic disturbance characteristics - Limits and methods of measurement. - EN 61000-4-2 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 2: Electrostatic discharge immunity test. - EN 61000-4-3 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 3: Radiated, radio-frequency, electromagnetic field immunity test. - EN 61000-4-4 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 4: Electrical fast transient/burst immunity test. - EN 61000-4-5 - Electromagnetic compatibility (EMC) - Part 4: Testing and measurement techniques - Section 5: Surge immunity test. - EN 61000-4-6 - Electromagnetic compatibility (EMC)- Part 4: Testing and measurement techniques - Section 6: Immunity to conducted disturbances, induced by radio-frequency fields.
Mechanical Construction	- EN 60529 e UL 50

Notes

- 1) Motor power is orientative, valid for standard WEG Motors of IV poles. The correct sizing must be done according to the nominal current of the motor used, which must be less than or equal to the rated output current of the inverter;
- 2) Braking resistor is not included;
- 3) Considering minimum line impedance of 1%;
- 4) For more information, refer to the user manual of CFW500;
- 5) All images are merely illustrative.
- 6) For operation with switching frequency above nominal, apply derating to the output current (refer to the user manual).



MASSA BRUTA/GROSS WEIGHT				MASSA LÍQUIDA/NET WEIGHT				ESCALA/SCALE		3:1							
				JEANCG						27.08.2020		00					
ECM		LOC		RESUMO DE MODIFICAÇÕES SUMMARY OF MODIFICATIONS				EXECUTADO EXECUTED		VERIFICADO CHECKED		LIBERADO RELEASED		DATA DATE		VER	
EXEC		JEANCG				CFW500 MEC B / SIZE B IP66				10007006683 00							
VERIF/CHECKED																	
LIBER/RELEASED																	
DT LIBER/REL DT		27.08.2020		WAU		JARAGUÁ DO SUL		DESENV. DRIVES SERIADOS		FOL/SHEET		1 / 1					